

8.7 POPULATION

8.7.1 Population attributes

8.7.2 Population growth

8.7.3 Lifehistory variation

In nature, animals of any species which live as isolated single individuals are very rare. Majority of them live in groups, in a well defined geographical area. They share or compete for similar resources and interbreed to produce fertile offspring. Such a group of individuals is called a population. Though the term interbreeding implies sexual reproduction, generally a group of individuals resulting from even asexual reproduction is also considered a '**population**' in ecological terms. e.g. all the frogs living in a lake, rats in an abandoned dwelling, teakwood trees in a forest, bacteria on a *culture plate*, lotus plants in a pond and Paramecia in a pond.

It is a well known fact that an individual organism is the one that has to adjust to a changed environment. But it is also important to know that Natural Selection operates at the level of population in evolving the desired traits. Population ecology is, therefore, an important area of ecology because it links ecology to population genetics and evolution.

8.7.1 Population Attributes

A population has certain attributes that an individual organism does not exhibit.

- 1. Natality (Birth rate) and Mortality (Death rate):** An individual may have birth and death, but a population has **birth rate (natality)** and **death rate (mortality)**. Natality is the number of new individuals produced in a unit time, per unit population. Mortality is the number of deaths that occurred in a unit time, per unit population. In a population these rates refer to *per capita* (per individual) births and deaths, respectively. The rate refers to change in numbers (increase or decrease) with respect to members of the population in unit time. Here is an example. If, in a pond there were 20 frogs last year and through reproduction 8 new frogs are added, taking the current population to 28, we calculate the birth rate as $8/20 = 0.4$ offspring per frog per year. If 4 individuals in a laboratory population of 40 fruit-flies died during a specified time interval, say a week, the death rate in the population during that period is $4/40 = 0.1$ individuals, per fruit-fly per week.
- 2. Immigration and Emigration:** **Immigration** is the number of individuals of the same species that have come into the habitat/population from elsewhere/another population during the time period under consideration. Immigration contributes to an increase in the population size.
Emigration is the number of individuals of population which left the habitat/population and gone elsewhere during the time period under consideration. Due to emigration the size of the population decreases.

3. **Sex ratio:** Another characteristic that influences the growth of a population is the *sex ratio*. An individual is either a male or a female, but a population has a sex ratio (e.g., 60 percent of the population are males and 40 percent population are females).
4. **Age distribution:** A population at any given time is composed of individuals of different ages. If the age distribution (percent individuals of a given age or age group) is plotted for the population, the resulting structure is called an '**age pyramid**'. For human population, the age pyramids generally show age distribution of males and females in a combined diagram. The shape of the pyramids reflect the 'status' of the population whether it is growing, stable or declining.

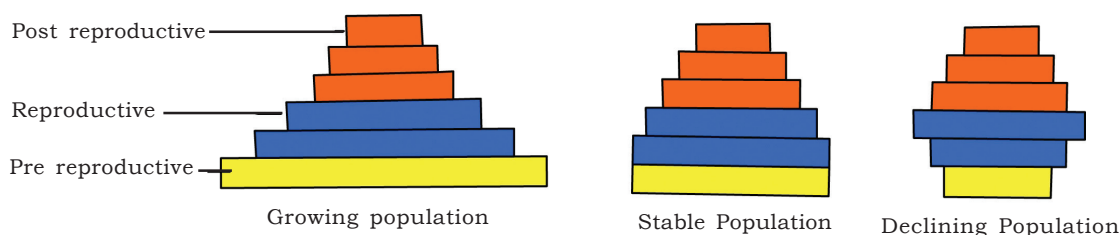


Fig. 8.21 Age pyramids

5. **Population size or Population density:** Population size, more technically called population density (designated as N) is the number of individuals per unit area (on land) or unit volume (in water).

Population density need not necessarily be measured in numbers only. Generally the most appropriate measure of population density is the total number. But in some cases it is either meaningless or difficult to determine. For example: Imagine there are 200 *Parthenium* plants and only a single huge banyan tree covering a large part in a particular area. If a statement is given that the population density of banyan is low relative to that of *Parthenium*, it amounts to underestimating the enormous role of the Banyan tree in that community. In such cases, the percent cover or biomass is a more meaningful measure of the population size. Total number is again not an easily adoptable measure if the population is huge and counting is impossible or very time-consuming. Sometimes, for certain ecological investigations, there is no need to know the absolute population densities; relative densities serve the purpose equally well. For instance, counting the number of fish caught per trap is good enough to measure its total population density in the lake. We mostly estimate population sizes indirectly, without actually counting them or seeing them. The tiger census in our National Parks and Tiger Reserves is often based on 'pug marks' and 'fecal pellets'. The size of the population provides a lot of information on the status of the species in the habitat. In the investigation of certain ecological processes in a population, change in the population size can be taken as an important parameter. e.g., the outcome of inter-specific competition, the impact of a predator, the effect of a pesticide application, etc., In nature, population size could be as low as <10 (Siberian

cranes at Bharatpur wetlands (Keoladeoghana National Park) in any year) or go into millions (*Chlamydomonas* in a pond).

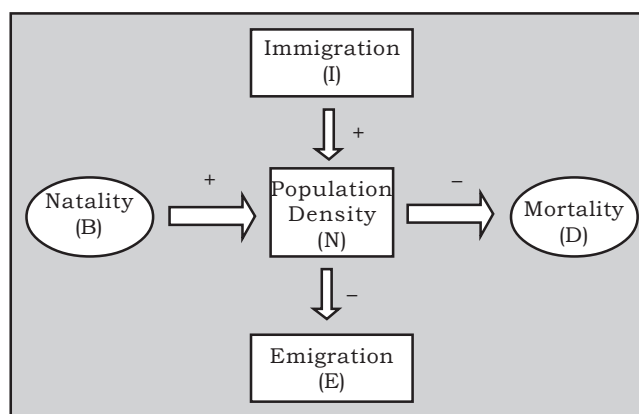
8.7.2. Population Growth

The size of a population for any species is not a static parameter. It keeps changing in time, depending on various factors including availability of food, 'predation pressure' and '**environmental resistance**'. These changes in population density provide information whether it is '*flourishing* or *declining*'. Whatever might be the ultimate reasons, the density of a population in a given habitat during a given period, fluctuates due to changes in four basic processes/ factors. **Natality** and **immigration** contribute to an increase in population density where as **mortality** and **emigration** lead to a decrease.

- (i) **Natality** refers to the number of births during a given period in the population that are added to the initial number of individuals.
- (ii) **Mortality** is the number of deaths in the population during a given period.
- (iii) **Immigration** is the number of individuals of the same species that have come into the habitat/ population from elsewhere/ another population during the time period under consideration.
- (iv) **Emigration** is the number of individuals of the population who left the habitat / population, during the time period under consideration.

So, if "**N**" is the population density at a given time '**t**', its density at the time **t + 1** is $N_{t+1} = N_t + \{(B + I) - (D + E)\}$

It can be seen from the above equation that population density will increase if the number of births plus the number of immigrants (**B + I**) is more than the number of deaths plus the number of emigrants (**D + E**), otherwise it will decrease. Under normal conditions, births and deaths are the most important factors influencing population density, the other two factors assuming importance only under special conditions. For instance, if a new habitat is just being colonised, immigration may contribute more significantly to the population growth than the factors affecting population density birth rate.



Growth Models

Does the growth of a population with time show any specific and predictable pattern? Uncontrolled growth of human population and resultant problems in our country are seriously concerning our planners. Therefore naturally we are curious to know whether different animal populations in nature behave the same way or

show some restraints on growth. Perhaps we can learn a lesson or two from nature on how to control population growth.

- (i) **Exponential growth:** Resource (**food** and **space**) availability is essential for the unhindered growth of a population. Ideally, when resources in the habitat are unlimited (environmental resistance is almost nil), each species has the potential ability to grow in number, as Darwin observed while developing his theory of Natural Selection. In such conditions, the population grows in an '**exponential**' or '**geometric**' fashion. If in a population of size **N**, the birth rates (not total number but *per capita* births) are represented as **b** and death rates (again, *per capita* death rates) as **d**, then the increase or decrease in **N** (**dN/dt**) during a unit time period **t** will be:

$$\frac{dN}{dt} = (b - d)N$$

Let

$$(b - d) = r, \text{ then}$$

$$\frac{dN}{dt} = rN$$

The '**r**' in this equation is called the '**intrinsic rate of natural increase**' and is a very important parameter chosen for assessing impacts of any biotic or abiotic factor on population growth.

To give you some idea about the range of **r** values, for the Norway rat, the **r** is 0.015, and for the flour beetle- it is 0.12. In 1981, the **r** value for human population in India was **0.0205**. To calculate 'r', we need to know the rates of birth and death.

The above equation describes the exponential or geometric growth pattern of a population (see figure) and results in a J-shaped curve when we plot **N** in relation to time. Using the principles of calculus, we can derive the integral form of the exponential growth equation as

$$N_t = N_0 e^{rt} \quad \text{where}$$

N_t = Population density after the time **t** ;

N₀ = Population density at the time zero (initial time);

r = Intrinsic rate of natural increase

e = The base of natural logarithms (2.71828)

Any species growing exponentially under unlimited resource conditions can reach enormous population densities in a short time. Darwin showed how even a slow growing animal like elephant could reach enormous numbers in the absence of '**natural checks**'.

- (ii) **Logistic growth:** No population of any species in nature has at its disposal unlimited resources to permit exponential growth. This leads to competition between individuals for limited resources. Eventually, the 'fittest' individual will

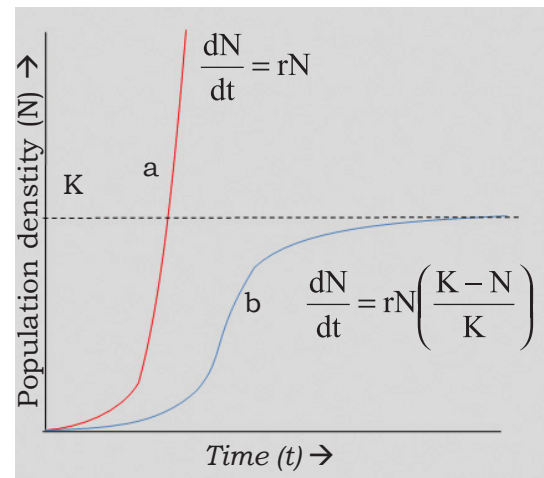
survive and reproduce. The governments of many countries have also realised this fact and introduced various control measures to limit growth of human population. In nature, any given habitat will have enough resources to support a maximum possible number, beyond which no further growth is possible. Let us call this limit as nature's **carrying capacity (K)** for that species in that habitat.

A population growing in a habitat with limited resources show initially a **lag phase**, followed by phases of **acceleration** and **deceleration** and finally an **asymptote/ stability**, when the population density reaches the carrying capacity. A plot of **N** in relation to time (**t**) results in a **sigmoid curve**. This type of population growth is called **Verhulst-Pearl Logistic Growth** (see figure) and is described by the following equation:

$$\frac{dN}{dt} = rN \left[\frac{K - N}{K} \right]$$

Where

- N** = Population density at the time t
r = Intrinsic rate of natural increase
K = Carrying capacity



Population growth curve

- 'a' : when resources are not limiting the growth plot is exponential.
 'b' : when resources are limiting the growth. plot is logistic,
 'K' : is carrying capacity

Since resources for growth for most animal populations are fixed and become limiting sooner or later, the logistic growth model is considered a more realistic one.

Population Growth curves: Curve 'a' shows the *exponential growth* in population when resources are unlimited; Curve 'b' shows the *logistic growth* in population when resources are a limiting factor where **K** is the *carrying capacity*.

8.7.3 Life history variation

Populations tend to maximise their reproductive fitness in their habitats. This is also called Darwinian fitness (high '**r**' value). Under a particular set of selection pressures, organisms evolve towards the most efficient reproductive strategy. Some organisms breed only once in their lifetime (e.g. Pacific salmon fish, bamboo plant) while others breed many times during their lifetime (e.g. most birds and mammals). Some produce a large number of small-sized offspring (e.g. Oysters, pelagic fishes) while others produce a small number of large-sized offspring (e.g. birds, mammals). The question is – which traits are desirable for maximising fitness? Currently this is an important area of research in ecology. Ecologists suggest that life history traits of organisms have evolved in relation to the constraints imposed by the abiotic and biotic components of the habitat in which they live.